Corvinus University of Budapest

Institute of Economics

Structural Determinants of Non-Communicable Disease Mortality Across Countries

Project 2 Research

Author: Meshkovski Nikola

BSc Applied Economics

Year 2025

Project 2 Instructor: Dr. Ferenci Tamás

TABLE OF CONTENTS

1. INTRODUCTION ..... - 2 -

STUDY BACKGROUND AND MOTIVATION ..... - 2 -

RESEARCH HYPOTHESIS ..... - 2 -

2. LITERATURE REVIEW ..... - 3 -

3. THE DATA ..... - 4 -

DATA SOURCE ..... - 4 -

DATA SAMPLE AND INITIAL DATA CLEANING ..... - 4 -

CONTENT OF VARIABLES ..... - 5 -

STATISTICAL OVERVIEW ..... - 7 -

4. METHODOLOGY ..... - 8 -

ANALYTICAL TECHNIQUES ..... - 8 -

ESTIMATION EQUATION ..... - 9 -

5. RESULTS ..... - 9 -

6. SUMMARY ..... - 13 -

7. REFERENCES ..... - 14 -

8. USAGE OF AI IN THE PAPER ..... - 14 -

1. INTRODUCTION

Non-communicable diseases (NCDs) are a group of medical conditions that are not transmissible directly from one person to another. Unlike infectious diseases, NCDs are typically chronic in nature, progress slowly, and require long-term management. The most common and impactful NCDs include cardiovascular diseases, cancers, chronic respiratory diseases, and diabetes. These four categories are the leading causes of premature death globally.

STUDY BACKGROUND AND MOTIVATION

In 2021, NCDs were responsible for over 41 million deaths, representing 74% of all non-pandemic-related deaths globally. Of these, approximately 17 million occurred among individuals younger than 70 years, with 86% of such premature deaths concentrated in low- and middle-income countries. This is the grim global face of non-communicable diseases, a slow-moving yet relentless crisis that accounts for nearly three-quarters of all deaths worldwide (WHO, 2021). These numbers don't merely reflect individual misfortune, they reveal structural weaknesses driven by a combination of demographic, environmental, and behavioral factors. Urbanization, aging populations, and the globalization of unhealthy lifestyles, such as poor diets, physical inactivity, tobacco use, and harmful alcohol consumption, have all contributed to the increasing prevalence of NCDs worldwide (PAHO, 2025). While the burden of NCDs has prompted increasing policy attention, mortality rates still vary significantly across countries. Such disparities suggest that broader structural determinants—such as national income, healthcare investment, pollution, and socioeconomic inequality are key in shaping population health outcomes (Zhou et al., 2020). For example, financial hardship due to out-of-pocket health payments or poor access to environmental health services has been linked to increased risk of NCD-related mortality (Murphy et al., 2020). Despite some evidence supporting the role of healthcare spending in reducing mortality, studies have produced mixed findings, particularly in settings where increases in spending may be offset by inefficiencies or environmental harms (Eckelman & Sherman, 2016). This study aims to explain that gap by integrating these dimensions through a multi-factor country-level analysis.

RESEARCH HYPOTHESIS

The central hypothesis of this study is that country-level mortality from non-communicable diseases (NCDs) is shaped by a complex interplay of structural determinants, including

economic capacity, behavioural risk factors, and environmental stressors. Specifically, this thesis proposes that higher alcohol consumption, lower national income, elevated pollution levels, and limited access to clean water are jointly associated with higher premature NCD mortality across countries. To examine this, we implemented two empirical strategies: a multivariable linear regression model to assess the average relationship between these variables and mortality, and a quantile regression model to capture how these effects vary across the distribution of NCD mortality rates. The methodology is discussed in detail further on. This dual approach aims to illuminate not only whether structural factors matter, but how their influence differs between countries with low, moderate, and high NCD burdens.

2. Literature Review

To contextualize our analysis of structural determinants of non-communicable disease (NCD) mortality, we draw on a selection of studies that employ diverse statistical approaches to assess the economic, environmental, and biological drivers of NCDs. Murphy et al. (2020) used data from over 180,000 individuals in 18 countries, applying regression analysis adjusted for demographic and health system factors. Their models estimated the association between NCD presence and outcomes such as healthcare utilization, out-of-pocket spending, and catastrophic health expenditure. By comparing households with and without NCDs, they identified significantly higher financial burdens for NCD households, especially in low-income settings. This highlights how economic strain contributes to health vulnerability. Ezzati et al. (2018) conducted a comparative analysis using WHO Global Health Estimates and NCD Risk Factor Collaboration data, focusing on low- and middle-income tropical countries. They examined age-standardized mortality rates across regions and linked them to determinants such as poverty, early-life nutrition, and environmental exposure. Though the study is descriptive in nature, it synthesizes epidemiological and demographic data to support a structural, multi-factor explanation of elevated NCD mortality. Wang and Wang (2020) employed multilevel modeling (MLM) techniques to estimate NCD mortality using WHO data from 176 countries. They grouped countries by economic development using a two-level Geotree classification and accounted for nested effects at both the group and country levels. Their results demonstrated that while overall NCD mortality declines with national wealth, the proportion of deaths due to NCDs increases, showing the importance of modeling heterogeneity. Eckelman and Sherman (2016) introduced a different dimension by applying an Economic Input-Output Life Cycle Assessment (EIOLCA) model to U.S. health expenditure and emissions data from 2003 to 2013.

This method linked financial flows in healthcare to sector-wide environmental impacts, estimating that the U.S. healthcare system contributed 10% of national GHG emissions and nearly 470,000 disability-adjusted life years (DALYs) lost. While not a traditional regression study, the EIOLCA provides a systems-level perspective on indirect consequences of health spending, complementing our environmental variables. Finally, López-Tenorio et al. (2024) conducted a comprehensive review of microbiome studies, including randomized clinical trials and mechanistic analyses. They synthesized results across intervention designs, such as fecal microbiota transplants (FMT), probiotic supplementation, and observational metabolite data, to explore the role of gut dysbiosis in NCD onset. Their work connects behavioral and environmental conditions to physiological health pathways, reinforcing our rationale for incorporating modifiable risk factors like alcohol and pollution in our models. Together, these studies not only inform the choice of variables in our analysis but also justify the application of both linear and distribution-sensitive models (like quantile regression) to better capture variation in NCD mortality across contexts.

3. The Data

Data Source

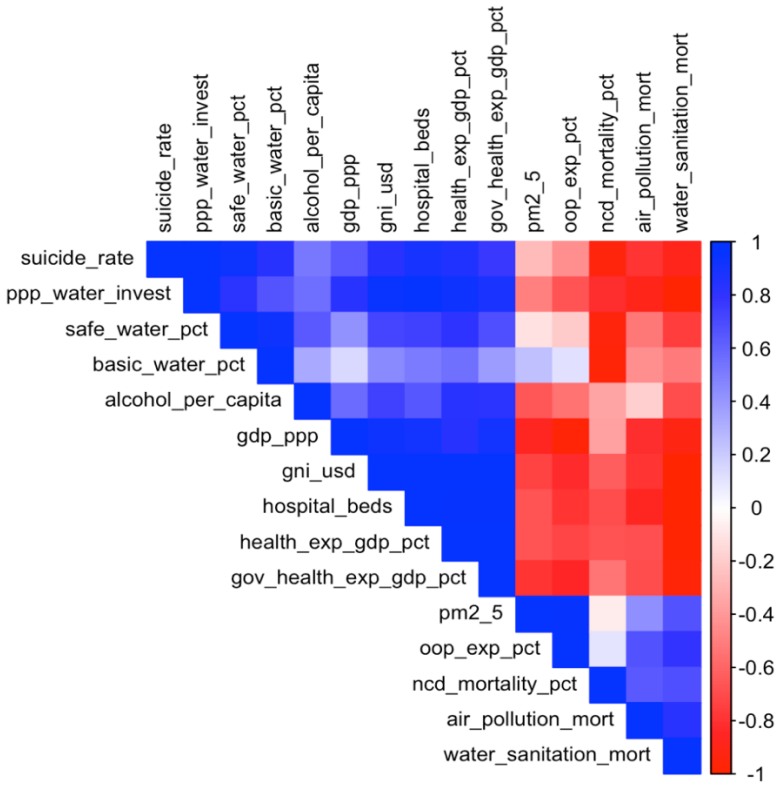
This study relies on country-level data primarily sourced from globally recognized and publicly accessible databases. The dataset includes a comprehensive set of economic, healthcare, and environmental indicators compiled from the World Bank World Development Indicators, the WHO Global Health Observatory, the Global Burden of Disease (IHME) project, and WHO/UNICEF JMP water and sanitation metrics. The analysis is restricted to data from 2019 to ensure temporal consistency across variables.

Data Sample and Initial Data Cleaning

For this study, the initial dataset comprised approximately 230 observations. The data represented a mixture of countries, income groups, and regional aggregates. Since our objective was to conduct a country-level cross-sectional analysis, all non-country observations—such as regional summaries (e.g., “World,” “Sub-Saharan Africa,” “Upper-middle income”), were excluded. In addition to filtering by geographic level, the final sample was restricted to observations with non-missing values for the key dependent variable, NCD mortality (%) among individuals aged 30–70. This ensured model validity and prevented inconsistency later

during regression analysis. The sample covers countries across varying levels of income and development, we ensured that the sample has a representative global scope aligned with the study's comparative approach. This overall refinement yielded a sample of 183 individual countries (observations) for analysis.

Figure 1: Correlation Heat Map of Data Sample Variables



Source: Author's own calculations based on data from the World Bank

Furthermore, Figure 1 presents a correlation heat map used to examine the strength of linear relationships between variables. This diagnostic step was critical in identifying and eliminating multicollinearity issues within the dataset. Variables exhibiting extremely high pairwise correlations (above 0.95) were removed to prevent redundancy and distortion in the regression analysis. As a result, the initial pool of over 20 candidate variables was refined to a final selection of seven key indicators used in the models.

CONTENT OF VARIABLES

The variables selected in this study are grounded in global public health and development priorities, capturing a broad spectrum of socio-environmental and economic determinants

associated with noncommunicable disease (NCD) mortality. Each variable has been selected based on its theoretical and empirical relevance, with several directly aligning with Sustainable Development Goals (SDGs), particularly SDG 3 (*Good Health and Well-being*), SDG 6 (*Clean Water and Sanitation*), and SDG 13 (*Climate Action*).

ncd_mortality_pct: This variable measures the probability (%) of dying between ages 30 and 70 from the four major NCD categories: cardiovascular diseases, cancer, diabetes, and chronic respiratory diseases. It reflects premature mortality and is directly associated with SDG Indicator 3.4.1, which aims to reduce by one-third premature mortality from NCDs through prevention and treatment. The indicator captures the cumulative risk profile faced by middle-aged adults and serves as the dependent variable in this analysis.

alcohol_per_capita: Defined as the total alcohol consumption in liters of pure alcohol per capita (aged 15+), this variable acts as a behavioral risk proxy for several NCDs. High levels of alcohol consumption are linked with liver cirrhosis, cancers, cardiovascular conditions, and increased injury risk. According to the World Health Organization (2025), reducing alcohol consumption is key to NCD control and aligns with SDG Target 3.5, which focuses on strengthening the prevention and treatment of substance abuse.

gni_usd_by1000: This variable reflects Gross National Income per capita in USD (scaled by 1,000 for interpretability). It captures a country's economic capacity, development status, and indirectly, access to health-promoting infrastructure and services. Although not tied to a specific SDG health indicator, GNI per capita is widely used across global development benchmarks and is central to the World Bank's income classification system, affecting health expenditure and service access.

suicide_rate: Measured per 100,000 population, this variable serves as a proxy for national mental health burden and societal stress. Suicide rates are a critical component of SDG Indicator 3.4.2, which seeks to reduce deaths from suicide and promote mental health and well-being. Mental health remains underrepresented in public health policy, and this variable bridges that gap in the context of broader NCD analysis.

pm2_5: This variable records the mean annual exposure to fine particulate matter (PM2.5, in $\mu g/m^3$), representing ambient air pollution. Long-term exposure is known to elevate the risk of cardiovascular and respiratory diseases. PM2.5 is indirectly tied to SDG 11.6.2 and SDG 3.9.1,

which track the health impacts of air quality and aim to reduce deaths caused by pollution. It serves as an environmental determinant in the model.

air_pollution_mort: This indicator measures the mortality rate per 100,000 attributed to both household and ambient air pollution. It captures the combined exposure effects and is part of SDG Indicator 3.9.1, targeting the reduction of deaths and illnesses from hazardous air. It strengthens the ecological dimension of the explanatory model, particularly for low- and middle-income countries where exposure remains high.

safe_water_pct: Representing the share of the population with access to safely managed drinking water services, this variable reflects environmental health infrastructure. Contaminated water is a vector for disease and indirectly contributes to chronic health outcomes through cumulative exposure. It aligns with SDG Indicator 6.1.1, emphasizing universal access to safe and affordable drinking water.

STATISTICAL OVERVIEW

Table 1: Descriptive Statistics of Key Variables

Variables	N	Mean	SD	Min	Median	Max
ncd_mortality_pct	183	19,8	7,7	7,3	19,9	50,8
alcohol_per_capita	182	5,4	4	0	4,9	17
gni_usd_by1000	177	22,3	22,7	0,8	14,1	121,4
suicide_rate	183	9,5	8	0,4	7,5	72,4
pm2_5	183	25,2	15,3	5,4	20,9	81
air_pollution_mort	183	100,8	78	7,4	83,4	305,1
safe_water_pct	117	68,3	30,3	6,1	74,6	100

Source: Author's own calculations based on data from the World Bank

Table 1 shows us the key descriptive statistics for the variables used in the analysis. NCD mortality rates vary widely, with a mean of 19.8% and a range from 7.3% to 50.8%. Alcohol consumption averages 5.4 liters per capita, while income levels (GNI per capita) range from $779 to $121,404. Environmental risks are also substantial, with PM2.5 exposure reaching up to 81 µg/m³, and air pollution mortality exceeding 300 deaths per 100,000 in some countries. Access to safe water remains uneven, with coverage as low as 6.1% in certain regions. These variations underscore the global inequality in structural health conditions.

4. METHODOLOGY

ANALYTICAL TECHNIQUES

To investigate the structural determinants of premature mortality from non-communicable diseases (NCDs), this study employs two primary estimation techniques: Ordinary Least Squares (OLS) regression and Quantile Regression (QR). The former estimates the average effect of explanatory variables on NCD mortality across countries, while the latter accounts for heterogeneity by estimating effects at different points in the conditional distribution of the dependent variable.

OLS is appropriate for understanding the general direction and magnitude of relationships, under the assumption of linearity and homoskedastic errors (Dawson et al., 2004; Hassard, 1991). However, because country-level NCD mortality distributions are likely to vary significantly, particularly between low- and high-burden settings, OLS alone may obscure tail dynamics.

Quantile regression was introduced to supplement the analysis. QR enables estimation of effects at the 25th, 50th (median), and 75th percentiles of the mortality distribution, capturing variations in how structural determinants operate across different outcome levels (Annual Reviews, 2021).

Model performance and parsimony were evaluated using Bayesian Information Criterion (BIC) rankings. BIC penalizes model complexity while balancing goodness-of-fit, with lower values indicating preferred models (Schwarz, 1978; Fox, 2025). Based on the BIC, the final model that is presented in the estimation equation was selected.

Multicollinearity was assessed using correlation matrices and Variance Inflation Factors (VIFs), with a strict threshold applied to exclude collinear predictors, meaning variables were flagged if they exhibited a VIF value that exceeded 5 (>5). VIF quantifies the inflation of coefficient variances due to interdependencies among independent variables (Investopedia, 2025; SCIRP, 2023).

ESTIMATION EQUATION

Model 1: OLS Regression Estimation Equation:

$$NCD\ Mortality_i = \beta_0 + \beta_1 \cdot Alcohol_i + \beta_2 \cdot GNI_i + \beta_3 \cdot Suicide_i + \beta_4 \cdot PM2.5_i + \beta_5 \cdot Air\ Pollution\ \text{Mortality}_i + \beta_6 \cdot Safe\ Water\ \text{Access}_i + \varepsilon_i$$

Model 2: Quantile Regression Estimation Equation:

$$Q_\tau\ (NCD\ Mortality_i) = \beta_{0,\tau} + \beta_{1,\tau} \cdot Alcohol_i + \beta_{2,\tau} \cdot GNI_i + \beta_{3,\tau} \cdot Suicide_i + \beta_{4,\tau} \cdot PM2.5_i + \beta_{5,\tau} \cdot \text{Air Pollution Mortality}_i + \beta_{6,\tau} \cdot \text{Safe Water Access}_i$$

- Q_τ (*) denotes the conditional quantile function for quantile $\tau \in \{0.25, 0.5, 0.75\}$.
- Coefficients $\beta_{j,\tau}$ vary across quantiles $j \in \{1, 2, 3, 4, 5, 6, 7\}$.

5. RESULTS

Table 2: Bayesian Identification Criterion (BIC) Model Comparison Table

Model	BIC	Rank
Model 5 – Full Socio-Enviro (GNI)	699,33	1
Model 6 – Full Socio-Enviro (GDP)	699,33	2
Model 9 – Log Spending + Core Drivers	1038,4	3
Model 7 – Reduced Socio-Enviro (No Water)	1046,04	4
Model 8 – Core Predictors Only	1072,17	5
Model 4 – Log-Linear Spending Model	1147,22	6
Model 2 – Spending + GDP	1149,33	7
Model 3 – Structural Factors	1156,04	8
Model 1 – Health Spending Only	1184,8	9

Source: Author's own calculations based on data from the World Bank

Table 3: Variance Inflation Factor (VIF) Results

Variable	Model	VIF	Flag
log_health_exp_usd	Model 4 – Log-Linear Spending Model	9,91	⚠️ > 5
log_health_exp_usd	Model 8 – Log Spending + Core Drivers	8,41	⚠️ > 5
log_gdp_ppp	Model 4 – Log-Linear Spending Model	8,23	⚠️ > 5
air_pollution_mort	Model 5 – Full Socio-Enviro (GNI)	4,8	
air_pollution_mort	Model 6 – Full Socio-Enviro (GDP)	4,8	
safe_water_pct	Model 5 – Full Socio-Enviro (GNI)	4,06	
safe_water_pct	Model 6 – Full Socio-Enviro (GDP)	4,06	

Source: Author's own calculations based on data from the World Bank

Model selection was guided by the Bayesian Information Criterion (BIC), with eight competing specifications evaluated. As shown in Table 2, Model 5, which includes alcohol consumption, GNI per capita, suicide rate, PM2.5 exposure, air pollution mortality, and safe water access, achieved the lowest BIC (699.33), indicating the best balance between model fit and parsimony. In addition to its statistical performance, Model 5 also passed multicollinearity diagnostics, as shown in Table 3. All Variance Inflation Factor (VIF) values fell well below the conservative threshold of 5, ranging from 1.32 to 4.80, suggesting no significant multicollinearity among the predictors. That said, several limitations must be acknowledged. The model is static, relying on predictors and outcomes from the same year (2019), which may overlook lagged relationships. Moreover, while the broad global sample enhances generalizability, it introduces greater heterogeneity in health systems and reporting standards. Lastly, while BIC is a useful model selection criterion, it may still lead to overfitting without external validation. Despite these limitations, Model 5 remains a strong, interpretable, and multidimensional explanatory model for cross-country variation in NCD mortality.

Table 4: OLS Regression Results for Final Model

Coefficients	Estimate	Std. Error	t value	Pr(>|t|)	Significance
(Intercept)	11,72	2,56	4,58	0,00	***
alcohol_per_capita	-0,15	0,13	-1,19	0,24	
gni_usd_by1000	-0,08	0,02	-3,36	0,00	**
suicide_rate	0,21	0,06	3,89	0,00	***
pm2_5	-0,10	0,04	-2,59	0,01	*
air_pollution_mort	0,08	0,01	7,20	0,00	***
safe_water_pct	0,04	0,03	1,47	0,15	
*When interpreting coefficients the logic of ceteris paribus is applied to all.					
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				Adjusted R^2 = 0.7056	

Source: Author's own calculations based on data from the World Bank

Observing Table 4, we can see that the baseline predicted NCD mortality when all explanatory variables are held at zero was 11.72 percentage points. We can also observe that alcohol consumption has a negative coefficient but is not statistically significant at any usual α levels (p=0.24). GNI per capita has a statistically significant negative effect on NCD mortality (p<0.5). A 1-unit increase in GNI (per $1,000) is associated with a 0.08 percentage point reduction in NCD mortality, highlighting the protective role of national income. Suicide rate shows a significant positive association with NCD mortality (p<0.001). A 1-unit increase in suicide mortality per 100,000 people corresponds to a 0.21 percentage point increase in the NCD death rate, suggesting that societal stress and mental health burdens may track broader health system performance. PM2.5 exposure is also significantly negatively associated with NCD mortality (p<0.05). A one-unit increase in fine particulate exposure contributes a decrease of 0.10 percentage points to the NCD mortality rate. Air pollution mortality has a positive and highly significant effect (p<0.001). Every additional 1-unit increase in Air pollution mortality per 100,000 people corresponds to a 0.08 percentage point rise in NCD mortality, underlining the impact of environmental conditions on chronic disease outcomes. Access to safe water is also not statistically significant at the 5% level (p=0.15).

Table 4: Quantile Regression Results for Final Model

Variable	Interpretation	25th Percentile	50th Percentile	75th Percentile
(Intercept)	Baseline NCD mortality at given quantile	4,074	4,365	12,173
alcohol_per_capita	Effect of alcohol consumption	0,093	-0,004	-0,135
gni_usd_by1000	Effect of GNI per capita	-0,093	-0,095	-0,079
suicide_rate	Effect of suicide rate	0,165	0,299	0,360
pm2_5	Effect of PM2.5 exposure	-0,053	-0,015	-0,062
air_pollution_mort	Effect of deaths from air pollution	0,089	0,090	0,087
safe_water_pct	Effect of access to safe water	0,071	0,081	0,023

Source: Author's own calculations based on data from the World Bank

To explore how explanatory variables influence NCD mortality across the distribution of NCDs, we estimated quantile regressions at the 25th, 50th, and 75th percentiles. This approach reveals differential effects not captured by a single OLS estimate. Alcohol consumption is positively associated at the lower quantile (0.093) but turns negative and more substantial at the 75th (−0.135), indicating that alcohol may exacerbate NCD risk, particularly in high-burden settings. GNI per capita consistently shows a negative effect across all quantiles, with the largest reduction observed at the 50th percentile (−0.095), confirming that higher income levels are associated with lower NCD mortality at both median and lower-risk country contexts. Suicide rate has a positive and increasing effect across quantiles, from 0.17 at the 25th percentile to 0.36 at the 75th. This suggests that in countries with higher baseline NCD mortality, mental health burdens play an even more prominent role. PM2.5 exhibits non-uniform effects: a moderately negative association at the 25th and 75th percentiles, but a smaller effect near the median. This suggests its influence may be more pronounced in countries with either very low or very high NCD burdens. Air pollution mortality remains consistently positive and stable across all quantiles (approx. 0.09), reinforcing its strong association with chronic disease outcomes regardless of mortality level. Access to safe water shows positive coefficients at all quantiles, highest at the median (0.081), but these are relatively modest. Overall, quantile regression results highlight important heterogeneity in predictor effects, suggesting that policy responses may need to be tailored based on a country's relative position in the NCD mortality distribution.

6. SUMMARY

This thesis set out to examine the structural drivers of premature mortality from non-communicable diseases (NCDs) across countries, motivated by growing global concern over uneven NCD outcomes and the need for more holistic, comparative analysis. Starting from a comprehensive dataset covering 183 countries, we focused on 2019 data to ensure cross-variable consistency and selected key socio-economic, environmental, and behavioral indicators as potential predictors. Our methodological approach involved both Ordinary Least Squares (OLS) and Quantile Regression. OLS provided a baseline understanding of average relationships, while quantile regression allowed us to investigate how these relationships differ across the NCD mortality distribution. Based on BIC model selection and multicollinearity testing, Model 5 was identified as the most suitable specification. This model included GNI per capita, suicide rate, PM2.5 exposure, air pollution mortality, alcohol consumption, and access to safe water. The results showed that economic capacity (GNI), environmental burden (PM2.5 and air pollution mortality), and mental health stressors (suicide rate) were significant determinants of NCD mortality. However, some variables such as alcohol consumption and water access were either not significant or varied in impact depending on the quantile, highlighting the heterogeneity of country contexts.

Our study is limited by its static, cross-sectional nature, reliance on non-age-standardized mortality data, and the absence of time-lagged predictors. These limitations point to opportunities for future work, such as panel data analysis, dynamic modeling, or integration of machine learning to enhance predictive scope. Despite these constraints, the study offers a robust framework for understanding the complex interplay of structural factors shaping NCD mortality outcomes worldwide.

7. References

Annual Reviews. (2021). *Quantile regression for survival data*. *Annual Review of Statistics and Its Application, 8*, 229–247. https://doi.org/10.1146/annurev-statistics-042720-125902

Dawson, B., & Hassard, T. (1991). *Understanding biostatistics*. Mosby-Year Book.

Eckelman, M. J., & Sherman, J. (2016). Environmental impacts of the U.S. health care system and effects on public health. *PLOS ONE, 11*(6), e0157014. https://doi.org/10.1371/journal.pone.0157014

Fox, J. (2025). *Applied regression analysis and generalized linear models* (4th ed.). SAGE Publications.

Investopedia. (2025). *Variance inflation factor (VIF)*. https://www.investopedia.com/terms/v/vif.asp

Murphy, A., Palafox, B., Walli-Attaei, M., Powell-Jackson, T., Rangarajan, S., Alhabib, K., ... & McKee, M. (2020). The household economic burden of non-communicable diseases in 18 countries. *BMJ Global Health, 5*(2), e002040. https://doi.org/10.1136/bmjgh-2019-002040

Pan American Health Organization. (2025). *Noncommunicable diseases*. https://www.paho.org/en/topics/noncommunicable-diseases

SCIRP. (2023). Dealing with multicollinearity in factor analysis. *Journal of Applied Mathematics, 14*(2), 101–109.

Schwarz, G. (1978). Estimating the dimension of a model. *Annals of Statistics, 6*(2), 461–464. https://doi.org/10.1214/aos/1176344136

World Bank. (2024). *World Development Indicators*. Retrieved from https://databank.worldbank.org/source/world-development-indicators

World Health Organization. (2021). *Noncommunicable diseases: Mortality*. https://www.who.int/data/gho/data/themes/topics/topic-details/GHO/ncd-mortality

World Health Organization. (2025). *Noncommunicable diseases: Fact sheets*. https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases

Zhou, M., Wang, H., Zeng, X., Yin, P., Zhu, J., Chen, W., ... & Naghavi, M. (2020). Modelling and prediction of global non-communicable diseases. *BMC Public Health, 20*, 822. https://doi.org/10.1186/s12889-020-08890-4

8. Usage of AI in the Paper

AI was primarily used to support refining mistakes during writing, check consistency across sections, and improve the overall clarity and coherence of the text. No AI methods were applied in the data analysis or model estimation processes. Perplexity AI was used as a tool to help with the search for articles, while ChatGPT was used for clarity and coherence.